

Lesson 4-4: Adding and Subtracting Rational Expressions

Math Objective	Understand that rational expressions form a system analogous to the system of rational numbers and use that understanding to add and subtract rational expressions.
Language Objective	Students will build rational expressions for a $d = rt$ word problem using a graphic organizer.
Essential Question	I can find the sum or difference of rational expressions.

CRITIQUE & EXPLAIN

Teo and Evander find the following exercise in their homework:

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{9}$$

A. Teo claims that a common denominator of the sum is $2 + 3 + 9 = 14$. Evander claims that it is $2 \cdot 3 \cdot 9 = 54$. Is either student correct? Explain why or why not.

B. Find the sum, explaining the method you use.

C. Construct Arguments Timothy states that the quickest way to find the sum of any two fractions with unlike denominators is to multiply their denominators to find a common denominator, and then rewrite each fraction with that denominator. Do you agree?

HABITS OF MIND Look for Relationships For two fractions with denominators 10 and x , when is $10x$ the least common multiple? When is $10x$ *not* the least common multiple?

What do you notice about the expression in the exercise?

How can there be more than one answer to the problem?

EXAMPLE 1 — Add or Subtract Rational Expressions with Like Denominators

When rational expressions have the same denominator, we combine the numerators and keep the denominator the same. This procedure is just like numerical fractions.

$$\frac{x+5}{x+4} + \frac{3}{x+4}$$

Since both fractions have the denominator $x+4$, we add the numerators:

$$\frac{x+5+3}{x+4} = \frac{x+8}{x+4}$$

Explain why it does not make sense to add the denominators?

Domain: The denominator $x+4$ cannot be zero, so

$$x \neq -4$$

TRY IT 1 Add or subtract the rational expressions. Then state the domain.

$\frac{4}{3x+1} + \frac{x+9}{3x+1}$	$\frac{2x-3}{x-7} - \frac{5}{x-7}$
Simplified form:	Simplified form:
Domain: $x \neq$ _____	Domain: $x \neq$ _____

EXAMPLE 2 — Find the LCD of Rational Expressions

To add or subtract rational expressions, we first need a **common denominator**. When denominators are factored, the **LCD** is the product of all unique factors.

Find the **LCD** of:

$$\frac{7}{x+4} \quad \text{and} \quad \frac{3}{(x+4)(x-6)}$$

Step 1 — Identify factors

- First denominator: $x + 4$
- Second denominator: $(x + 4)(x - 6)$

Step 2 — Combine all unique factors

$$\text{LCD: } (x + 4)(x - 6)$$

*Why didn't we include $x + 4$ twice?***TRY IT 2** Find the LCD of the following rational expressions:

$\frac{5}{x+2} \quad \text{and} \quad \frac{9}{(x+2)(x-5)}$	$\frac{3}{x-1} \quad \text{and} \quad \frac{5}{x^2-1}$ <i>Difference of squares!</i>
LCD:	LCD:

EXAMPLE 3 — Add or Subtract Rational Expressions with Unlike Denominators

When rational expressions have **different denominators**, the process is the same whether you are adding or subtracting:

1. **Find the LCD**
2. **Rewrite each fraction using the LCD**
3. **Combine the numerators** (add or subtract)
4. **State the domain**

Example A — Addition

$$\frac{5}{x+3} + \frac{x-2}{(x+3)(x-1)}$$

Step 1: LCD

$$(x + 3)(x - 1)$$

Step 2: Rewrite

$$\frac{5}{x+3} \cdot \frac{x-1}{x-1} = \frac{5(x-1)}{(x+3)(x-1)}$$

Why are we allowed to change fractions before adding them?

Second fraction already has the LCD.

Step 3: Add

$$\frac{5(x-1) + (x-2)}{(x+3)(x-1)}$$

Simplify numerator:

$$5x - 5 + x - 2 = 6x - 7$$

Final Answer

$$\frac{6x - 7}{(x+3)(x-1)}$$

Domain

$$x \neq -3, \quad x \neq 1$$

Example B — Subtraction

$$\frac{x+7}{x-3} - \frac{4}{x+2}$$

Step 1: LCD

$$(x-3)(x+2)$$

Step 2: Rewrite

$$\frac{x+7}{x-3} \cdot \frac{x+2}{x+2} = \frac{(x+7)(x+2)}{(x-3)(x+2)}$$

$$\frac{4}{x+2} \cdot \frac{x-3}{x-3} = \frac{4(x-3)}{(x-3)(x+2)}$$

Step 3: Subtract

$$\frac{(x+7)(x+2) - 4(x-3)}{(x-3)(x+2)}$$

Expand:

$$(x+7)(x+2) = x^2 + 9x + 14$$

$$4(x-3) = 4x - 12$$

Subtract:

$$x^2 + 9x + 14 - (4x - 12)$$

Be careful with the subtraction:

$$x^2 + 9x + 14 - 4x + 12 = x^2 + 5x + 26$$

Final Answer

$$\frac{x^2 + 5x + 26}{(x-3)(x+2)}$$

Domain

$$x \neq 3, \quad x \neq -2$$

What extra step(s) do we need here that we don't do with regular fractions?

How is the distributive property used when subtracting rational expressions?

TRY IT 3

Add these expressions $\frac{4}{x-5} + \frac{x+3}{(x-5)(x+1)}$ Simplified form: Domain: $x \neq$ _____, $x \neq$ _____	Subtract these expressions $\frac{x-8}{x+4} - \frac{6}{x+6}$ Simplified form: Domain: $x \neq$ _____, $x \neq$ _____
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EXAMPLE 5 — Use Rational Expressions to Solve Problems

Jorge drives his car to the mechanic, then he takes the commuter rail train back to his neighborhood. The average speed for the 10-mile trip is 15 miles per hour faster on the train. Find an expression for Jorge's total travel time. If he drove 30 mph, how long did this take?



	Distance	Rate	Time
Car			
Train			

How do you know you need to add these expressions to solve the problem?

What does each fraction represent in the context?

How is a table a useful tool in solving this problem?

TRY IT — 5 Jorge drives to visit his friends. His 20-mile trip to their house is congested with traffic, but it has cleared by the time he travels home. He can travel 5 miles per hour faster. Write an expression that represents Jorge's total travel time.

Select Exercises

3. Error Analysis A student added the rational expressions as follows: $\frac{5x}{x+7} + \frac{7}{x} = \frac{5x}{x+7} + \frac{7(7)}{x+7} = \frac{5x+49}{x+7}$ Describe and correct the error the student made. DOK2	
12. Generalize Explain how addition and subtraction of rational expressions is similar to and different from addition and subtraction of rational numbers. DOK2	
35. SAT/ACT What is the difference between $\frac{x}{9}$ and $\frac{x-y}{6}$? Ⓐ $\frac{5x-y}{18}$ Ⓑ $\frac{5x+y}{18}$ DOK1 Ⓒ $\frac{-x+3y}{18}$ Ⓓ $\frac{-x-3y}{18}$	
26. On Saturday morning, Ahmed decided to take a bike ride from one end of the 15-mile bike trail to the other end of the bike trail and back. His average speed the first half of the ride was 2 mph faster than his speed on the second half. Find an expression for Ahmed's total travel time. If his average speed for the first half of the ride was 12 mph, how long was Ahmed's bike ride? SEE EXAMPLE 5 DOK2	
31. Use Structure Aisha paddles a kayak 5 miles downstream at a rate 3 mph faster than the river's current. She then travels 4 miles back upstream at a rate 1 mph slower than the river's current. <i>Hint: Let x represent the rate of the river current.</i> a. Write and simplify an expression to represent the total time it takes Aisha to paddle the kayak 5 miles downstream and 4 miles upstream. b. If the rate of the river current, x, is 2 mph, how long was Aisha's entire kayak trip? DOK3	